Vectors I - HW1 (Introduction to Ratio Theorem)

Solutions

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Original practice questions. Total: 50 marks. Give exact answers unless otherwise stated.

Question 1 **[5 marks]**

The vectors $\mathbf{a}$ and $\mathbf{b}$ are given by $\mathbf{a}=2\mathbf{i}-\mathbf{j}+3\mathbf{k}$ and $\mathbf{b}=-\mathbf{i}+4\mathbf{j}$.

(a) Write $\mathbf{a}$ and $\mathbf{b}$ as column vectors and find $\mathbf{a}+2\mathbf{b}$. [2]

(b) Find $\mathbf{c}$ such that $2\mathbf{a}-3\mathbf{b}+\mathbf{c}=\mathbf{0}$. [2]

(c) Points P and Q have position vectors $\mathbf{a}+2\mathbf{b}$ and $\mathbf{c}$ respectively. Find $\overrightarrow{PQ}$. [1]

Solution

(a) $\mathbf{a}=\begin{pmatrix}2\\-1\\3\end{pmatrix}$, $\mathbf{b}=\begin{pmatrix}-1\\4\\0\end{pmatrix}$, $\mathbf{a}+2\mathbf{b}=\begin{pmatrix}2\\-1\\3\end{pmatrix}+2\begin{pmatrix}-1\\4\\0\end{pmatrix}=\begin{pmatrix}0\\7\\3\end{pmatrix}$.

(b) $\mathbf{c}=3\mathbf{b}-2\mathbf{a}=\begin{pmatrix}-3\\12\\0\end{pmatrix}-\begin{pmatrix}4\\-2\\6\end{pmatrix}=\begin{pmatrix}-7\\14\\-6\end{pmatrix}$.

(c) $\overrightarrow{PQ}=\overrightarrow{OQ}-\overrightarrow{OP}=\begin{pmatrix}-7\\14\\-6\end{pmatrix}-\begin{pmatrix}0\\7\\3\end{pmatrix}=\begin{pmatrix}-7\\7\\-9\end{pmatrix}$.

Question 2 **[6 marks]**

The points A and B have coordinates $(1,-2,2)$ and $(3,1,-4)$ respectively.

(a) Find a unit vector that is parallel to AB. [2]

(b) Find a vector of length 21 parallel to AB. [2]

(c) A point P is such that $\overrightarrow{AP}$ has magnitude 14 and the same direction as $\overrightarrow{AB}$. Find the coordinates of P and state the order of the three points along their common straight path. [2]

Solution

(a)
$$\overrightarrow{AB}=\begin{pmatrix}3\\1\\-4\end{pmatrix}-\begin{pmatrix}1\\-2\\2\end{pmatrix}=\begin{pmatrix}2\\3\\-6\end{pmatrix}, \qquad |\overrightarrow{AB}|=\sqrt{4+9+36}=7.$$

One such unit vector is $\frac{1}{7}\begin{pmatrix}2\\3\\-6\end{pmatrix}$. Its negative is also a valid answer.

(b) One such vector is $21\left(\frac{1}{7}\begin{pmatrix}2\\3\\-6\end{pmatrix}\right)=\boxed{\begin{pmatrix}6\\9\\-18\end{pmatrix}}$. Its negative is also a valid answer.

(c)
$$\overrightarrow{AP}=14\left(\frac{1}{7}\begin{pmatrix}2\\3\\-6\end{pmatrix}\right)=\begin{pmatrix}4\\6\\-12\end{pmatrix}, \qquad \overrightarrow{OP}=\overrightarrow{OA}+\overrightarrow{AP}=\begin{pmatrix}5\\4\\-10\end{pmatrix}.$$

Thus $\boxed{P(5,4,-10)}$. Since $\overrightarrow{AP}=2\overrightarrow{AB}$, the order is A, B, P.

Question 3 **[6 marks]**

The vectors $\mathbf{p}=\begin{pmatrix}6\\k\\-3\end{pmatrix}$ and $\mathbf{q}=\begin{pmatrix}2\\-4\\m\end{pmatrix}$ are parallel, where k and m are real constants.

(a) Find k and m. [3]

(b) State whether $\mathbf{p}$ and $\mathbf{q}$ have the same or opposite directions, and find $|\mathbf{p}|:|\mathbf{q}|$. [1]

(c) Find all vectors of magnitude $\sqrt{21}$ that are parallel to $\mathbf{p}$. [2]

Solution

(a) Since the vectors are parallel, $\mathbf{p}=h\mathbf{q}$ for some non-zero real h. Comparing first components gives $6=2h$, so $h=3$.

$$k=3(-4)=-12, \qquad -3=3m \quad \Longrightarrow \quad \boxed{k=-12,\ m=-1}.$$

(b) Since $\mathbf{p}=3\mathbf{q}$, they have the same direction and $\boxed{|\mathbf{p}|:|\mathbf{q}|=3:1}$.

(c) $|\mathbf{q}|=\sqrt{2^2+(-4)^2+(-1)^2}=\sqrt{21}$. Both directions must be included, so all such vectors are

$$\boxed{\pm\begin{pmatrix}2\\-4\\-1\end{pmatrix}}.$$

Question 4 **[6 marks]**

The points A, B and C have coordinates $(1,2,-1)$, $(3,1,2)$ and $(7,-1,8)$ respectively.

(a) Show that A, B and C are collinear, and find $AB:BC$. [3]

(b) The point $R(u,-2,v)$ lies on the same straight path as A, B and C. Find u and v. [3]

Solution

(a)

$$\overrightarrow{AB}=\begin{pmatrix}2\\-1\\3\end{pmatrix}, \qquad \overrightarrow{BC}=\begin{pmatrix}4\\-2\\6\end{pmatrix}=2\overrightarrow{AB}.$$

Thus $\overrightarrow{AB}$ and $\overrightarrow{BC}$ are parallel and have the common point B, so A, B, C are collinear. Taking magnitudes gives $\boxed{AB:BC=1:2}$.

(b) Collinearity gives $\overrightarrow{AR}=s\overrightarrow{AB}$ for some real s, so

$$\begin{pmatrix}u-1\\-4\\v+1\end{pmatrix}=s\begin{pmatrix}2\\-1\\3\end{pmatrix}.$$

The second components give $s=4$. Hence $u-1=8$ and $v+1=12$, giving $\boxed{u=9,\ v=11}$.

Question 5 **[8 marks]**

The points A and B have coordinates $(2,-1,3)$ and $(-4,5,0)$ respectively. A point M divides AB internally in the ratio $AM:MB=2:1$. A point N lies on AB produced beyond B and satisfies $AN:NB=5:2$.

(a) Find the position vector of M. [2]

(b) Find the position vector of N. [3]

(c) State the order of the four points A, B, M and N, and find $MB:BN$. [3]

Solution

(a) By the ratio theorem,

$$\overrightarrow{OM}=\frac{\overrightarrow{OA}+2\overrightarrow{OB}}{3}=\frac{1}{3}\left[\begin{pmatrix}2\\-1\\3\end{pmatrix}+2\begin{pmatrix}-4\\5\\0\end{pmatrix}\right]=\boxed{\begin{pmatrix}-2\\3\\1\end{pmatrix}}.$$

(b) Since $AN=AB+BN$ and $AN:BN=5:2$, $AB:BN=3:2$. Hence $\overrightarrow{BN}=\frac{2}{3}\overrightarrow{AB}$, and

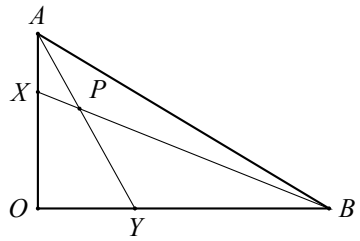
$$\overrightarrow{ON}=\overrightarrow{OB}+\frac{2}{3}(\overrightarrow{OB}-\overrightarrow{OA})=\frac{5\overrightarrow{OB}-2\overrightarrow{OA}}{3}=\boxed{\begin{pmatrix}-8\\9\\-2\end{pmatrix}}.$$

(c) The order is A, M, B, N. Since $MB=\frac{1}{3}AB$ and $BN=\frac{2}{3}AB$,

$$\boxed{MB:BN=1:2}.$$

Question 6 **[10 marks]**

Relative to the origin O, the points A and B have non-zero, non-parallel position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively. The point X lies on OA with $OX:XA=2:1$, and the point Y lies on OB with $OY:YB=1:2$. The line segments AY and BX intersect at P.



(a) By expressing $\overrightarrow{OP}$ in two ways using the ratio theorem, find $\overrightarrow{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. [6]

(b) Find $AP:PY$ and $BP:PX$. [2]

(c) The point Z divides AB internally with $AZ:ZB=1:4$. Show that O, P, Z are collinear and find $OP:PZ$. [2]

Solution

(a) First, $\overrightarrow{OX}=\frac{2}{3}\mathbf{a}$ and $\overrightarrow{OY}=\frac{1}{3}\mathbf{b}$. Let $AP:PY=s:(1-s)$ and $BP:PX=t:(1-t)$, where $0<s,t<1$. By the ratio theorem,

$$\overrightarrow{OP}=(1-s)\overrightarrow{OA}+s\overrightarrow{OY}=(1-s)\mathbf{a}+\frac{s}{3}\mathbf{b},$$
$$\overrightarrow{OP}=(1-t)\overrightarrow{OB}+t\overrightarrow{OX}=\frac{2t}{3}\mathbf{a}+(1-t)\mathbf{b}.$$

Since $\mathbf{a}$ and $\mathbf{b}$ are non-parallel, comparing coefficients gives

$$1-s=\frac{2t}{3}, \qquad \frac{s}{3}=1-t.$$

The second equation gives $s=3-3t$. Substituting into the first,

$$-2+3t=\frac{2t}{3} \quad\Longrightarrow\quad 7t=6 \quad\Longrightarrow\quad t=\frac{6}{7}, \quad s=\frac{3}{7}.$$

Consequently,

$$\boxed{\overrightarrow{OP}=\frac{4}{7}\mathbf{a}+\frac{1}{7}\mathbf{b}}.$$

(b) $\boxed{AP:PY=3:4, \qquad BP:PX=6:1}$.

(c) By the ratio theorem,

$$\overrightarrow{OZ}=\frac{4\mathbf{a}+\mathbf{b}}{5}, \qquad \overrightarrow{OP}=\frac{4\mathbf{a}+\mathbf{b}}{7}=\frac{5}{7}\overrightarrow{OZ}.$$

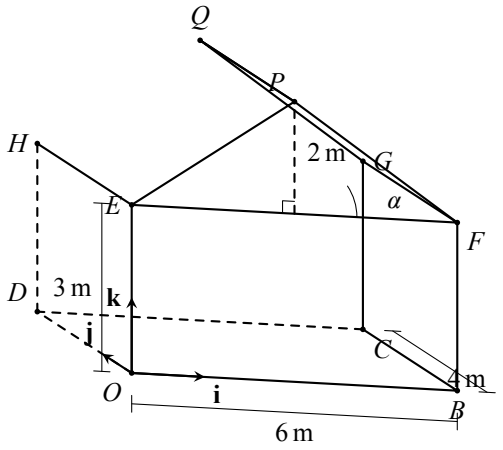
The two vectors are parallel and have the common point O, so O, P, Z are collinear. Since $0<\frac{5}{7}<1$, P is between O and Z; hence $\boxed{OP:PZ=5:2}$.

Question 7 **[6 marks]**

A solid house consists of a cuboid $OBCDEFGH$ and a triangular prism roof $EFPHGQ$, as shown. The base $OBCD$ is horizontal. The vertices E, F, G and H are directly above O, B, C and D respectively.

The front-left corner O is the origin. The positive **i**, **j** and **k** directions are along OB, OD and OE respectively, and each unit represents 1 metre. It is given that $OB = 6$ m, $OD = 4$ m and $OE = 3$ m.

The ridge PQ is parallel to OD. The front gable EFP and back gable HGQ are congruent isosceles triangles, with P vertically above the midpoint of EF and Q vertically above the midpoint of HG. Each sloping roof face makes an angle α with the horizontal, where $\tan\alpha = \frac{2}{3}$; the vertical roof rise is 2 m.



All lengths are in metres. The diagram is not drawn to scale.

Find the coordinates of all the vertices of the house $(O, B, C, D, \ldots)$.

Solution

The positive coordinate directions are along OB, OD and OE. Hence the four base vertices have $z = 0$, the front vertices have $y = 0$, and the back vertices have $y = 4$.

The walls are 3 m high, so E, F, G, H all have $z = 3$. The roof is symmetric across the middle of its 6 m width, so the ridge has $x = \frac{6}{2} = 3$.

Writing h for the vertical roof rise,

$$\tan\alpha = \frac{h}{6/2} = \frac{2}{3} \quad \Rightarrow \quad h = 2\text{ m}.$$

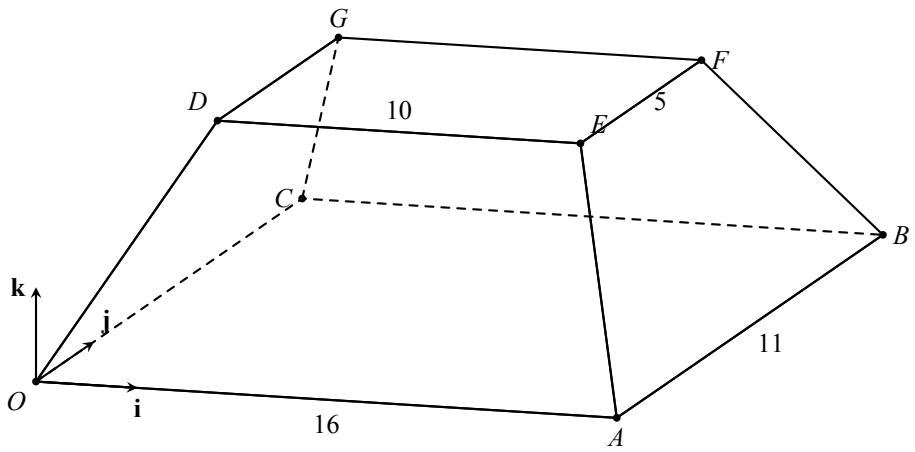
This agrees with the labelled rise. The ridge is at height $3 + 2 = 5$ m. Therefore all ten vertices are

$$\begin{array}{|ll|}\hline O=(0,0,0), & B=(6,0,0),\\ C=(6,4,0), & D=(0,4,0),\\ E=(0,0,3), & F=(6,0,3),\\ G=(6,4,3), & H=(0,4,3),\\ P=(3,0,5), & Q=(3,4,5).\\ \hline\end{array}$$

Question 8 **[3 marks]**

The diagram shows a container with a horizontal rectangular base $OABC$, where $OA = 16$ cm and $AB = 11$ cm. The top of the container $DEFG$ is also a horizontal rectangle, where $DE = 10$ cm and $EF = 5$ cm. The four sloping faces (for example $OAED$) of the container, each a trapezium, are inclined at the same angle to the horizontal, and the vertical distance between the base and the top is 6 cm.

The point O is taken as the origin $(0, 0, 0)$, and perpendicular unit vectors **i**, **j**, **k** are such that **i** is parallel to OA, **j** is parallel to OC, and **k** is vertical.



All lengths are in centimetres. The diagram is not drawn to scale.

Find the coordinates of D.

Solution

Since the four sloping faces are inclined at the same angle to the horizontal, the top rectangle $DEFG$ is centred above the base rectangle $OABC$.

The horizontal inset from each base edge to the corresponding top edge is

$$\frac{OA - DE}{2} = \frac{16 - 10}{2} = 3, \qquad \frac{AB - EF}{2} = \frac{11 - 5}{2} = 3.$$

The top lies 6 cm above the base, so the z-coordinate of every top vertex is 6. Taking D as the front-left vertex of the top,

$$\boxed{D = (3, 3, 6)}.$$

END OF HOMEWORK